



INDEX PROCESSES AND MARKOV PROCESSES

► **Reversible:** the index process equals reduces to a simple random walk on $\{0, \dots, N\}$ with reflective boundaries and transition probabilities:

► **Non-Reversible:** the index process equals reduces to a persistent random walk on $\{0, \dots, N\}$ with reflective boundaries and transition probabilities:

► Otherwise:

$$I_{t+1} = I_t, \quad \epsilon_{t+1} \sim \text{uniform}\{-1, 1\}$$

► If $0 \leq I_t + \epsilon_t \leq N$ then with probability s_{I_t, ϵ_t} the index process satisfies:

$$I_{t+1} = I_t + \epsilon_t, \quad \epsilon_{t+1} \sim \text{uniform}\{-1, 1\}$$

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$$I_{t+1} = I_t + \epsilon_t, \quad \epsilon_{t+1} = \epsilon_t$$

► Otherwise let

$$I_{t+1} = I_t, \quad \epsilon_{t+1} = -\epsilon_t$$

► Under the ELE the Index process (I_t^m, ϵ_t^m) is Markovian chain with initial condition:

$$(I_0^m, \epsilon_0^m) = (m, (-1)^m)$$



Stationary with respect to $\text{Uniform}\{0, \dots, N\} \times \text{Uniform}\{-1, 1\}$

INDEX PROCESS IS A MARKOV PROCESS

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- ▶ **Reversible:** the index process equals reduces to a simple random walk on $\{0, \dots, N\}$ with reflective boundaries and transition probabilities:

- ▶ If $0 \leq I_t + \epsilon_t \leq N$ then with probability s_{I_t, ϵ_t} the index process satisfies:

$$I_{t+1} = I_t + \epsilon_t, \quad \epsilon_{t+1} \sim \text{uniform}\{-1, 1\}$$

- ▶ Otherwise:

$$I_{t+1} = I_t, \quad \epsilon_{t+1} \sim \text{uniform}\{-1, 1\}$$

- ▶ **Non-Reversible:** the index process equals reduces to a persistent random walk on $\{0, \dots, N\}$ with reflective boundaries and transition probabilities:

- ▶ If $0 \leq I_t + \epsilon_t \leq N$ then with probability s_{I_t, ϵ_t} the index process satisfies:

$$I_{t+1} = I_t + \epsilon_t, \quad \epsilon_{t+1} = \epsilon_t$$

- ▶ Otherwise let

$$I_{t+1} = I_t, \quad \epsilon_{t+1} = -\epsilon_t$$

ROUND-TRIP TIMES